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Sociality predicts enhanced auditory thresholds and binaural hearing processing across rodent species

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Sociality has evolved repeatedly in rodents, yet whether sociality has shaped auditory processing remains unclear. Here, we used auditory brainstem responses (ABRs) to examine the relationship between sociality and auditory processing in nine wild-caught rodent species that are not closely related evolutionarily. Phylogenetic comparative analyses revealed that sociality is correlated with enhanced auditory processing: group-living and monogamous species generally exhibited lower auditory thresholds for tone and click stimuli than solitary species. Group-living species also showed higher relative amplitudes of the binaural interaction component (BIC) across tested interaural time differences (ITDs) than solitary species. In contrast, BIC normalized latencies did not differ among social groups across tested ITDs. We also found no evidence of differences among social groups in monaural ABR interpeak latencies or amplitude ratios of ABR waves I and IV with increasing sound intensities. Together, these findings indicate that sociality is associated with heightened auditory sensitivity in rodents and suggest that the evolution of complex social behavior may have favored enhanced auditory systems that facilitate effective social communication.

Animal behavior has long been recognized as capable of reshaping organisms' selective pressures, a phenomenon particularly evident in social communication settings where signals mediate interactions among individuals. Acoustic communication, in particular, plays a central role in animal social dynamics by mediating key biological behaviors, such as territorial defense, collective movement, group cohesion, mate attraction, predator avoidance, and the transmission of information about individual identity or emotional states[1–6]. Because behavior both reflects and shapes the ecological challenges animals face, recurrent social interactions create consistent selective pressures on the sensory and signaling systems involved. As a result of these diverse and crucial functions, acoustic signals are under strong selective pressures that shape both their structural and perceptual properties[7].

For decades, comparative studies have primarily focused on the sender side of the mammalian acoustic communication, demonstrating that interspecific variation in body size, sexual selection pressures, and formant-related cues are major drivers of vocal signal diversity[8–11]. However, effective communication also depends on the receiver's auditory system, which natural selection is expected to tune to the acoustic properties of conspecific signals to ensure efficient information transfer[12–14]. This co-evolutionary relationship between behavior, signal production, and sensory perception highlights how behavioral demands can structure the evolution of communication systems. Although comparative studies of ear morphology have provided important insights into auditory abilities across vertebrates[15–18], especially in rare or non-model species, direct physiological measures of auditory processing remain limited for most vocalizing vertebrates. Consequently, the extent to which vocal signal complexity influences auditory sensitivity is still poorly understood.

Rodents make up nearly 40% of all described mammals and are an excellent model system for behavioral and auditory research owing to their extensive variation in social structure[19]. However, comparative analyses of rodent hearing, especially among wild-caught species, remain scarce. To date, behavioral and physiological audiograms have been reported for more than 25 rodent species, with a disproportionate focus on a limited set of model organisms, most notably the laboratory house mouse (*Mus musculus*) and the Norway rat (*Rattus norvegicus*)[20]. These studies reveal substantial interspecific variation in auditory processing, with some species showing peak sensitivity to low-frequency, while others exceed human capabilities in detecting higher frequencies[21]. For instance, the Zambian mole-rat (*Fukomys amatus*), Ansell mole-rat (*Fukomys anselli*), and the common mole-rat (*Cryptomys hottentotus*) exhibit the lowest frequency hearing thresholds between 0.8 to 1.4 kHz, whereas the house mouse (*Mus musculus*), hispid pocket mouse (*Chaetodipus hispidus*), prairie vole (*Microtus ochrogaster*), and members of the genus *Peromyscus* show the

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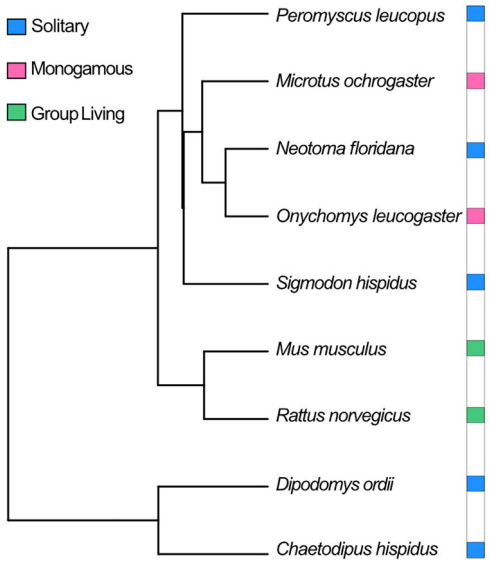


Fig. 1 | Phylogenetic relationships and social organization of wild-caught rodents included in the study. Phylogeny of the wild-caught rodents used in this study, reconstructed from the mammalian tree generated by Upham and colleagues[94] based on the Basic Local Alignment Search Tool (BLAST) of National Center for Biotechnology Information (NCBI) nucleotide database. Blue squares represent solitary-living rodents, pink squares represent Monogamous rodents, and green squares represent Group-living rodents.

greatest auditory sensitivity at frequencies between 8 to 46 kHz[22–29]. Despite these findings, direct cross-species comparisons, specifically species that exhibit different form of social organizations, remain challenging owing to methodological differences that often lead to inconsistent results. Therefore, standardized comparative studies on rodent auditory perception are increasingly necessary to better understand variation in auditory acuity across species and social groups.

Sociality has evolved multiple times in tetrapods, and its relationship with acoustic communication has been extensively investigated[30–33]. Across primates, birds, and cetaceans, increased complexity, particularly in group-living species is correlated with greater vocal repertoire diversity and signal complexity[11,34,35]. For example, primates tend to expand their vocal repertoires with increasing group size to support a broader range of social and ecological interactions[36]. A classic example is the ring-tailed lemur (*Lemur catta*) which produces high frequency, modulated calls to alert dispersed group members to predators or convey urgency[37]. Rodents provide a complementary and highly tractable system for investigating this relationship, given their widespread reliance on acoustic communication. Across species, rodents produce both audible and ultrasonic vocalizations that mediate key behaviors, including alarm signaling, mate attraction, and parent-offspring interactions[38–41]. For instance, ground-dwelling sciurid rodents produce alarm calls that encode predator type and urgency[42–45], while house mice (*Mus musculus*) emit ultrasonic calls in social and reproductive contexts[38,46]. Additionally, substantial differences have been reported in the complexity of vocal signals across rodents varying in social structure, with social rodents generally exhibiting greater vocal repertoire diversity compared to both monogamous and solitary rodents[47–49]. For instance, the highly social house mouse (*Mus musculus*) produces 15 call types, while monogamous prairie voles (*Microtus ochrogaster*) and solitary Key Largo woodrats (*Neotoma floridana smalli*) emit approximately 14 and 4 call types, respectively[47–49]. Despite these well-documented differences in vocal behavior, it remains unclear whether variation in social structure is associated with corresponding differences in auditory processing across rodent species. Consequently, comparative studies of auditory sensitivity across species with differing social structure are critical to elucidate how sociality shapes the evolution of sensory processing.

The main goal of this study was to examine the relationship between sociality and auditory processing using nine wild-caught rodent species that are not closely related evolutionarily (Fig. 1, Table 1). We measured auditory sensitivity through auditory brainstem responses (ABRs), including ABR thresholds, click thresholds (Fig. 2A), interpeak latencies, amplitude ratios, binaural normalized latencies and binaural relative amplitudes across social groups (solitary, monogamous, and group-living). We hypothesized that group-living and monogamous rodents would exhibit more sensitive auditory systems characterized by lower ABR thresholds across relevant frequencies and lower click thresholds across socially relevant frequencies compared to solitary species. We also predicted that group-living species would show higher amplitude, shorter latency responses to click stimuli, as well as enhanced binaural responses amplitudes and latencies relative to other social groups.

Results

Phylogenetic heritability

Molecular identification based on cytochrome c oxidase subunit I (COI) sequences confirmed the species identity of all 90 individuals representing the nine species examined (Fig. 1, Table 1). Phylogenetic comparative analyses revealed generally weak phylogenetic signals in auditory thresholds and ABR wave characteristics across social groups, as reflected by low estimates of phylogenetic heritability. Phylogenetic heritability was estimated at 0.23 [95% Credible Interval (95% CrI): 0.00 to 0.61] for ABR thresholds, 0.06 (95% CrI: 0.00 to 0.47) for click thresholds, 0.13 (95% CrI: 0.03 to 0.38) for interpeak latency, 0.08 (95% CrI: 0.01 to 0.31) for amplitude ratio, 0.10 (95% CrI: 0.01 to 0.39) for BIC normalized latency, and 0.02 (95% CrI: 0.00 to 0.13) for BIC relative amplitude.

Hearing thresholds differ across social groups

All social groups exhibited the greatest hearing sensitivity (i.e., lowest ABR thresholds) between 8 and 24 kHz (Fig. 2B, Supplementary Fig. 1). Bayesian phylogenetic generalized linear mixed models (PGLMMs) revealed that group-living species had lower ABR thresholds than solitary species (Posterior mean difference (PM) = 12.34; 95% CrI: 2.53 to 22.06; Bayesian Markov Chain Monte Carlo *P*-value (P_{MCMC}) = 0.021). Similarly, monogamous rodents had lower ABR thresholds than solitary species (PM = 10.14; 95% CrI: 2.77 to 17.23; P_{MCMC} = 0.012). Bayesian pairwise contrast showed that group-living species displayed lower ABR thresholds than solitary species at 1 kHz (PM = 12.34; 95% CrI: 2.52 to 22.06; P_{MCMC} = 0.021), 2 kHz (PM = 12.79; 95% CrI: 3.27 to 21.96; P_{MCMC} = 0.014), 4 kHz (PM = 10.15; 95% CrI: 0.64 to 19.23; P_{MCMC} = 0.040), 8 kHz (PM = 12.17; 95% CrI: 2.27 to 21.33; P_{MCMC} = 0.020), and 16 kHz (PM = 11.31; 95% CrI: 1.51 to 20.54; P_{MCMC} = 0.028). However, group-living and solitary species exhibited similar hearing thresholds at 24 kHz (PM = 8.49; 95% CrI: −1.25 to 17.88; P_{MCMC} = 0.078), 32 kHz (PM = 7.55; 95% CrI: −1.98 to 16.63; P_{MCMC} = 0.106), 46 kHz (PM = 4.49; 95% CrI: -4.88 to 13.65; P_{MCMC} = 0.284), and 64 kHz (PM = 3.07; 95% CrI: −6.29 to 12.30; P_{MCMC} = 0.456).

Monogamous species showed lower ABR thresholds than solitary species at 1 kHz (PM = 10.14; 95% CrI: 2.77 to 17.23; P_{MCMC} = 0.012), 2 kHz (PM = 12.99; 95% CrI: 5.54 to 20.06; P_{MCMC} = 0.004), 4 kHz (PM = 11.44; 95% CrI: 4.10 to 18.58; P_{MCMC} = 0.006), and 8 kHz (PM = 11.04; 95% CrI: 3.37 to 18.27; P_{MCMC} = 0.011). In contrast, no substantial differences were detected in ABR hearing thresholds between monogamous and solitary species at 16 kHz (PM = 7.27; 95% CrI: −0.19 to 14.12; P_{MCMC} = 0.056), 24 kHz (PM = 5.15; 95% CrI: −2.36 to 12.51; P_{MCMC} = 0.151), 32 kHz (PM = 0.77; 95% CrI: −6.87 to 7.77; P_{MCMC} = 0.808), 46 kHz (PM = 3.43; 95% CrI: −4.01 to 10.62; P_{MCMC} = 0.328), and 64 kHz (PM = 2.80; 95% CrI: −4.70 to 10.05; P_{MCMC} = 0.403). There was no evidence for a difference in ABR thresholds between monogamous and group-living species across tested frequencies (PM = 2.20; 95% CrI: −7.77 to 13.37; P_{MCMC} = 0.656).

Click thresholds were also substantially lower for group-living compared to solitary species (PM = 9.23; 95% CrI: 2.17 to 16.0; P_{MCMC} = 0.013;

Table 1 | Species characteristics and sampling information for wild-caught rodents included in the study

Scientific name	Social structure	Activity pattern	Age group	Body mass range (g)	Sample size (Juvenile, subadult, adult)	References
Mus musculus	Group-living	Nocturnal	Juvenile Subadult Adult	< 10 11-14 > 14	10 (2,3,5)	91,105,106
Rattus norvegicus	Group-living	Nocturnal/diurnal	Juvenile Subadult Adult	< 100 101-120 > 120	10 (3,1,6)	90,107,108
Microtus ochrogaster	Monogamous	Nocturnal/diurnal	Juvenile Subadult Adult	< 20 21-30 > 30	10 (1,2,7)	109–112
Onychomys leucogaster	Monogamous	Nocturnal	Juvenile Subadult Adult	< 17 18-25 > 25	10 (0,1,9)	89,113,114
Sigmodon hispidus	Solitary	Nocturnal/diurnal	Juvenile Subadult Adult	< 80 81-110 > 110	10 (3,3,4)	115–117
Chaetodipus hispidus	Solitary	Nocturnal	Juvenile Subadult Adult	< 30 30-35 > 35	10 (6,0,4)	87,118,119
Peromyscus leucopus	Solitary	Nocturnal	Juvenile Subadult Adult	< 13 14-18 > 18	10 (1,3,6)	88,120–122
Dipodomys ordii	Solitary	Nocturnal	Juvenile Subadult Adult	< 40 41-52 > 52	10 (0,3,7)	86,123,124
Neotoma floridana	Solitary	Nocturnal	Juvenile Subadult Adult	< 100 101-140 > 140	10 (1,2,7)	75,125

Scientific name, social structure, activity pattern, age group, body mass range (g), sample size by age class (Juvenile, subadult, and adult), and corresponding references for each species included in the study.

Fig. 2C, see Supplementary Fig. 2 for species click thresholds). Likewise, monogamous species had lower click thresholds than solitary species (PM = 8.50; 95% CrI: 1.51 to 14.8; $P_{MCMC} = 0.011$). No substantial evidence for differences were detected in click thresholds between monogamous and group-living species (PM = 0.72; 95% CI: −7.70 to 9.57; $P_{MCMC} = 0.426$). These findings suggest that group-living and monogamous rodents both have lower ABR and click response thresholds than solitary rodents, specifically at lower frequencies.

No differences in ABR interpeak latency and amplitude ratio across social groups

To assess differences in monaural ABR wave characteristics, we calculated interpeak latency and amplitude ratio of ABR waves I and IV across tested intensities (90–50 dB SPL). Interpeak latency was measured as the difference between the peak latencies of wave I and wave IV at each intensity level for the left and right pinnae, while amplitude ratio was calculated by dividing the value of wave IV amplitude by the value of wave I amplitude for the right and left pinnae at each intensity. As expected, interpeak latency of wave I and wave IV decreased with increasing intensity (Fig. 3A, Supplementary Fig. 4A). However, our PGLMM indicated no significant differences in interpeak latency among social groups. Specifically, interpeak latency did not differ either between group-living and solitary species (PM = −0.08; 95% CrI: −0.45 to 0.27; $P_{MCMC} = 0.659$), or between group-living and monogamous species (PM = 0.09; 95% CrI: −0.29 to 0.49; $P_{MCMC} = 0.617$). We also found no evidence for a difference in interpeak latency of ABR wave I and IV between monogamous and solitary species (PM = −0.18; 95% CrI: −0.49 to 0.12; $P_{MCMC} = 0.249$).

Overall, amplitude ratio increased with increasing intensities, with wave I consistently showing larger values than wave IV across the tested intensity range (90–50 dB SPL; Fig. 3B; Supplementary Fig. 4B). However, the PGLMM revealed no significant differences in the wave IV/I amplitude ratio between group-living and solitary species (PM: −0.12; 95% CrI: −0.33 to 0.09; $P_{MCMC} = 0.253$). Similarly, no significant differences were observed in wave IV/I amplitude ratio either between group-living and monogamous species (PM: 0.002; 95% CrI: −0.21 to 0.24; $P_{MCMC} = 0.998$), or between monogamous and solitary species (PM: −0.12; 95% CrI: −0.31 to 0.07; $P_{MCMC} = 0.179$). Taken together, these results indicate no evidence of differences in monaural peak ABR wave I and wave IV across rodents differing in social behavior organization.

Binaural hearing differences across social groups

To quantify binaural hearing sensitivity for each social group, we extracted the peak latency and amplitude of the binaural ABR data at each ITD and transformed to BIC normalized latency and BIC relative amplitude with respect to ITD at 0 ms[50]. The average BIC normalized latency and BIC relative amplitude were used to compare binaural hearing sensitivity as a function of ITD across social groups. PGLMM revealed no evidence for differences in BIC normalized latency between group-living and solitary species across tested ITDs (PM: 0.28; 95% CrI: −0.04 to 0.62; $P_{MCMC} = 0.083$, Fig. 3C). Similarly, no substantial differences were detected in BIC normalized latency between monogamous and solitary species across tested ITDs (PM: 0.26; 95% CrI: −0.02 to 0.52; $P_{MCMC} = 0.062$). There was also no evidence for a substantial difference in BIC normalized latency between group-living and monogamous species (PM: 0.03; 95% CrI: −0.33 to 0.38; $P_{MCMC} = 0.889$). Despite no overall evidence for a substantial difference in BIC normalized latency between monogamous and solitary species, the Bayesian pairwise contrast showed that monogamous species had slightly faster BIC normalized latency than solitary species at 1.0 ms and 1.5 ms ITDs (Supplementary Fig. 5A; Supplementary Table 1). Likewise, group-living slightly exhibited faster BIC normalized latency than solitary species at 1.5 ms ITD (Supplementary Table 1).

Lastly, the PGLMM model revealed that group-living species had higher BIC relative amplitude than solitary species (PM: −13.23; 95% CrI: −23.67 to −1.96; $P_{MCMC} = 0.029$, Fig. 3D). Bayesian pairwise contrast showed that group-living species exhibited higher BIC relative amplitude

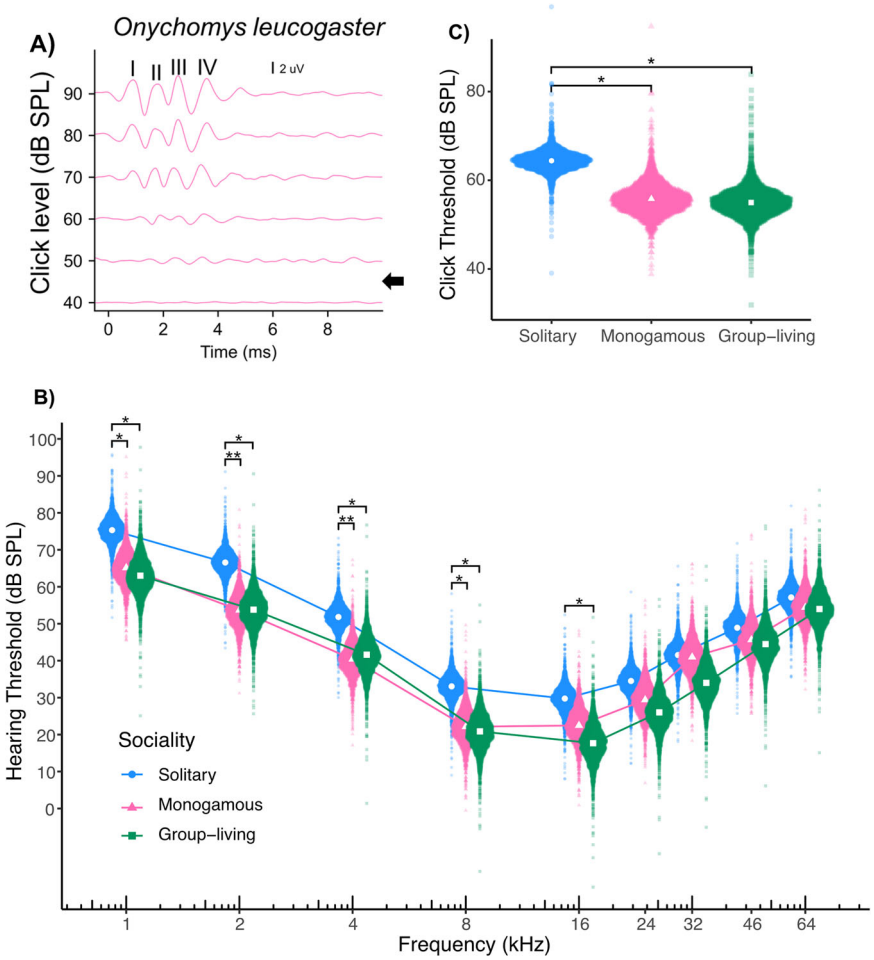


Fig. 2 | Sociality is associated with enhanced auditory brainstem response sensitivity across wild-caught rodents. A ABR traces evoked by click stimuli from a representative male monogamous species (*Onychomys leucogaster*) included in this study. Waves I – IV are labeled at the 90 dB SPL, and the arrow between 50- and 40-dB SPL indicated the click thresholds (See Supplementary Fig. 3 for an example of each tested species). **B** Mean auditory brainstem response thresholds in solitary, monogamous, and group-living wild-caught rodents across tested frequencies ($n = 90$, 10 individuals per species). **C** Click response thresholds across social groups (solitary = blue, monogamous = pink, group-living = green). White circle (solitary), white triangle (Monogamous), and white square (Group-living) represent the mean value for each group ($n = 90$, 10 individuals per species). PGLMM showed Significant differences in click thresholds and ABR thresholds among social groups [(ABR thresholds: Group-living vs solitary ($P_{MCMC} = 0.021$); Monogamous vs solitary ($P_{MCMC} = 0.012$); Click thresholds: Group-living vs solitary ($P_{MCMC} = 0.013$); Monogamous vs solitary ($P_{MCMC} = 0.011$, two-sided Bayesian pairwise comparisons of posterior distributions were used for multiple comparisons)]. Asterisks denote Bayesian statistical support. ** $P_{MCMC} < 0.01$, * $P_{MCMC} < 0.05$ (effects considered significant when 95% credible interval exclude zero).

than solitary species at 0.5 ms ITD (PM: −13.23; 95% CrI: −23.69 to −1.96; $P_{MCMC} = 0.025$), 1.0 ms ITD (PM: −17.17; 95% CrI: −27.96 to -6.09; $P_{MCMC} = 0.005$), 1.5 ms ITD (PM: −19.64; 95% CrI: −30.003 to −9.007; $P_{MCMC} = 0.002$), and 2.0 ms ITD (PM: −10.69; 95% CrI: −21.17 to −0.26; $P_{MCMC} = 0.045$). There were, however, no significant overall differences in BIC relative amplitude either between monogamous and solitary species (PM: −3.98; 95% CrI: −14.01 to 6.46; $P_{MCMC} = 0.436$) or between monogamous and group-living species (PM: −9.25; 95% CrI: −21.89 to 3.82; $P_{MCMC} = 0.133$), illustrating similar BIC relative amplitude among these social groups. Although no significant differences were detected in overall BIC relative amplitude between monogamous and solitary species, the Bayesian pairwise contrast showed that monogamous species had higher relative amplitude than solitary species at 2.0 ms ITD (Supplementary Table 2). Likewise, group-living had higher relative amplitude than monogamous species at 1.5 ms ITD (Supplementary Fig. 5B; Supplementary Table 2). These findings indicate that group-living rodents have more sensitive BIC relative amplitude than solitary rodents across ITDs.

Discussion

In this study, phylogenetic comparative analyses were used to investigate the relationship between sociality and auditory processing across wild-caught rodent species, representing three social organizations. All species showed the greatest auditory sensitivity between 8 and 24 kHz (Fig. 2B), consistent with previous reports showing that small-headed mammals of similar size exhibit the lowest auditory sensitivity around this frequency range[24–26,28,51–53]. Our analyses revealed that group-living and monogamous rodents exhibited lower hearing thresholds than solitary rodents. Similarly, we found that group-living rodents exhibited higher BIC relative amplitude responses than solitary living rodents. In contrast, we detected no significant differences in monaural ABR wave interpeak latency and amplitude ratio among social groups. There was also no evidence for a substantial difference in BIC normalized latency among social groups. Overall, these results indicate that rodents' auditory processing varies systematically with social organization, with group-living rodents generally exhibiting enhanced auditory sensitivity, lower click thresholds, and higher binaural amplitudes than solitary rodents. However, these results should be interpreted with care owing to the effective sample size for comparative inference being limited to nine species and to low phylogenetic signals detected in auditory thresholds and ABR wave characteristics measured.

Our results demonstrated that group-living and monogamous species tend to exhibit lower ABR thresholds than solitary species, with differences on the order of 10-20 dB SPL. However, statistically supported differences were restricted to specific frequency ranges, occurring between 1-8 kHz for monogamous versus solitary species and 1–16 kHz for group-living versus solitary species. For instance, group-living and monogamous rodents detected 1–4 kHz sounds up to 15 dB quieter than solitary rodents, with similar trends observed at higher frequencies (16–64 kHz, 10 dB SPL quieter). Consistent patterns were also observed for click evoked thresholds, with group-living and monogamous rodents generally exhibiting lower click thresholds than solitary rodents. These findings suggest that sociality is consistently associated with enhanced auditory sensitivity in rodents, particularly within low and mid-frequency ranges. Notably, this pattern parallels results in primates, where species living in larger groups exhibit improved auditory sensitivity and broader high-frequency hearing ranges[54]. Together, these cross-taxonomic parallels suggest that recurrent social interactions may impose similar evolutionary pressures on auditory systems

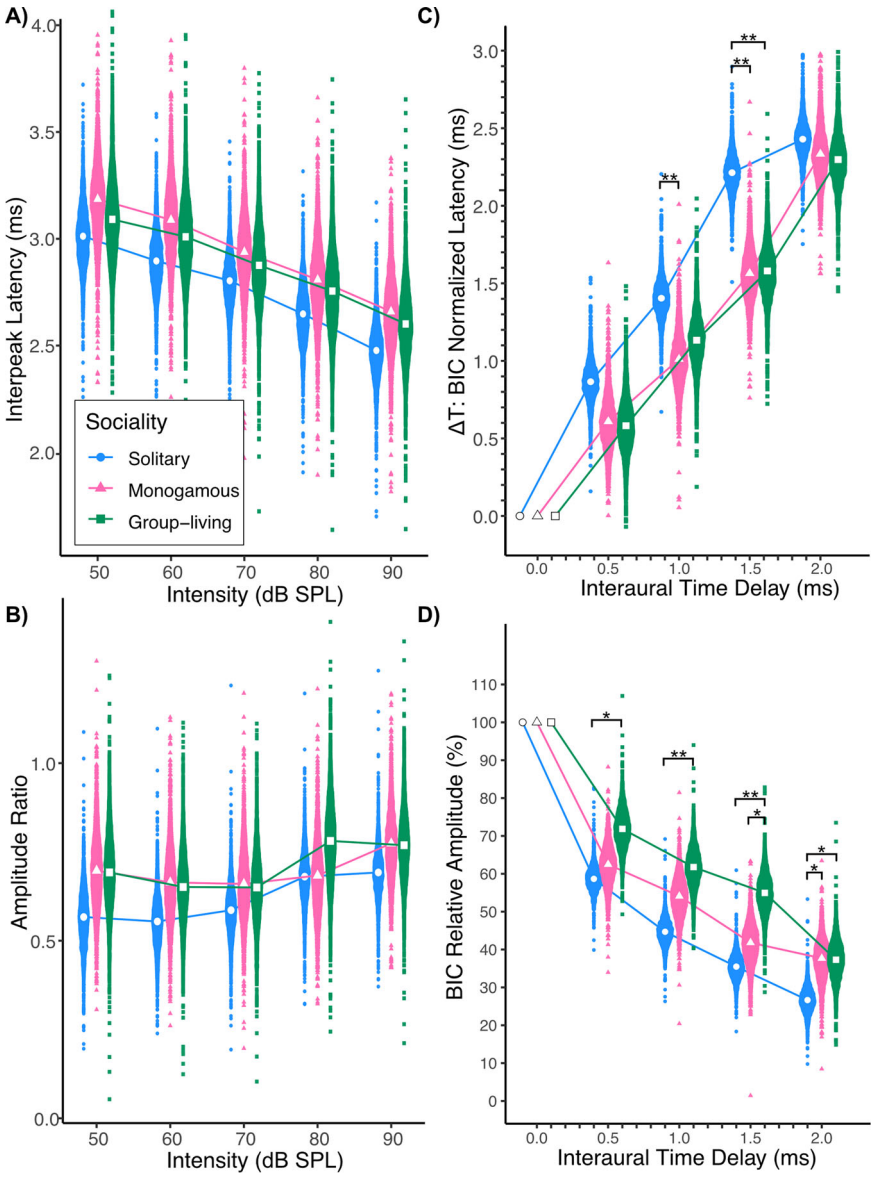


Fig. 3 | Monaural and binaural auditory brainstem response properties across social groups. A Mean interpeak latency of ABR waves I and IV determined with clicks of different intensities (blue represents solitary species, pink represents monogamous species, and green represents group-living species, $n = 90$, 10 individuals per species). **B** Mean ABR waves IV/I amplitude ratio determined with clicks of different intensities ($n = 90$, 10 individuals per species). No significant differences were detected in interpeak latency and amplitude ratio among social groups (PGLMM: All $P_{MCMC} > 0.05$). **C** Mean BIC normalized latency in solitary, monogamous, and group-living wild-caught rodents across tested ITD ($n = 90$, 10 individuals per species). **D** Mean BIC relative amplitude across groups ($n = 90$, 10 individuals per species). PGLMM showed significant differences in BIC relative amplitude between group-living and solitary groups ($P_{MCMC} = 0.029$, two-sided Bayesian pairwise comparisons of posterior distributions were used for multiple comparisons). White circle (solitary), white triangle (Monogamous), and white square (Group-living) represent the mean value for each group. Asterisks denote Bayesian statistical support. ** $P_{MCMC} < 0.01$, * $P_{MCMC} < 0.05$ (effects considered significant when 95% credible interval exclude zero).

across rodents tested in this study. Accordingly, our data support our first hypothesis that species engaging in more frequent or complex social communication evolve more sensitive auditory mechanisms, particularly across socially relevant frequency ranges, although this relationship appears frequency-dependent and warrants further investigation.

The social drive hypothesis posits that sociality can drive the evolution of improved perceptual and communicative abilities[54]. This hypothesis has been well supported in rodents, birds and primates, where species living in larger or more complex social groups tend to produce more diverse vocal repertoires, exhibit greater auditory sensitivity, and increase high frequency limits[11,34,54,55]. For example, in African striped mice (*Rhabdomys pumilio*), group size is associated with increased vocal diversity: individuals vocalize more frequently in the presence of familiar conspecifics and modulate their vocal repertoire when interacting with unfamiliar individuals[55]. Similarly, in primates, group size correlates with both vocal repertoire size and improved detection of quieter signals[11,54], consistent with the notion that greater social complexity imposes stronger selection on both the production and perception of acoustic information. The evolutionary pressures underlying this pattern are likely multifaceted. Firstly, group-living individuals must discriminate among a greater number of conspecific vocalizations across a wide range of acoustic contexts; and secondly, they may experience more variable environmental pressures in which efficient signal detection is of paramount importance for group cohesion, mate attraction, and predator avoidance. Enhanced auditory sensitivity in group-living and monogamous species, particularly at lower frequencies could therefore provide adaptive advantages by facilitating long distance communication, recognition of individual-specific vocal cues, and rapid detection of alarm calls. The similarity between our findings in rodents and those previously reported in primates suggests that sociality and auditory sensitivity may have coevolved repeatedly across mammals, revealing common selective pressures on communication systems in social species.

ABR-derived audiograms showed broadly similar shapes among social groups, with peak sensitivity occurring between 8 and 24 kHz. Although group-living and monogamous species had significantly lower ABR thresholds than solitary species at frequencies below 8 kHz, overall sensitivity in this range remained relatively poor (40 to 65 dB SPL). Small-headed mammals, like most members of the order Rodentia, are unable to extract reliable directional information from low-frequency sounds due to minimal interaural cues. Instead, they rely on high-frequency cues for sound localization, which likely explained their reduced sensitivity to low frequencies[56–59]. In addition, rodents are known to emit sonic and ultrasonic vocalizations spanning approximately from ~1 to 110 kHz, with peak

energy typically occurring between 10 and 35 kHz; however, these ranges are derived from studies of representative rodent taxa and not all species examined here have species-specific vocalization data available[49,60–63]. While all species were sensitive to frequencies that overlapped with rodents' ultrasonic calls, social species (both monogamous and group-living) showed greatest sensitivity for 8-16 kHz tones. Therefore, relatively better low-ABR sensitivity in group-living and monogamous species could have important social and ecological importance such as facilitating long-distance perception and transmission of vocal signals, whereas high-frequency call components may support short-range communication and thus may mediate social behaviors such as mother-pup interactions, group cohesion, courtship, territoriality, alarm calling, and aggression[52].

We further quantified interpeak latency and amplitude ratio of ABR waves I and IV across social groups with increasing stimulus intensities (50–90 dB SPL). In small-headed rodents, wave I typically occurs around 1.5–2.5 ms, likely reflecting neuronal activity in the auditory nerve and cochlear nucleus, whereas wave IV occurs at 4.5–5.5 ms, likely corresponding to input from the nucleus of the lateral lemniscus and/or early input to the inferior colliculus[64,65]. Consistent with previous studies[26,64,66], wave I-IV interpeak latency decreased with increasing stimulus intensity, with wave IV exhibiting the largest latency shift, while wave IV/I amplitude ratio increased with increasing intensity. Given that ABR wave metrics are sensitive to recording (e.g., electrodes placements, equipment characteristics) and physiological variables (e.g., body temperature), interpeak latency and amplitude ratio provide useful normalized measures for cross-species or group comparisons. Indeed, wave IV amplitude was consistently smaller than wave I (IV/I < 1.0) among social groups, which is consistent with prior reports suggesting that early ABR waves are typically larger than later waves[29,64,67]. Despite this, no evidence for substantial statistical differences were detected among social groups for either interpeak latency or amplitude ratio, suggesting that early auditory brainstem processing is conserved and not strongly shaped by sociality or shared evolutionary history in the nine rodent species examined here. However, differences in ABR wave morphology (I-IV) have been reported in species such as inbred house mice (*Mus musculus*), gerbils (*Meriones unguiculatus*), and guinea pigs (*Cavia porcellus*), and these differences are often driven by variation in central auditory organization and neural conduction, indicating that finer-scale specializations may not be captured by these metrics alone[64,68]. Future studies incorporating boarder taxonomic sampling and ecologically relevant stimuli will be necessary to continue exploring the potential association between sociality and auditory processing.

Our binaural ABR results are consistent with previous comparative studies showing that BIC normalized latency increases and BIC relative amplitude decreases with increasing ITD[28,50,69]. Prior studies have also demonstrated substantial interspecific variation in BIC DN1 waveform properties, with BIC relative amplitudes ranging from ~ 0.2 μV in humans to ~ 5 μV in other mammals such as guinea pigs, gerbils, cats, and *Peromyscus* species, whereas BIC normalized latencies typically range from ~3.7 to 6.8 ms across mammals[26,29,70–72]. In our study, BIC relative amplitudes and latencies at 0 ms ITD fell within these previously reported mammalian ranges (relative amplitude: 1.5 to 4.4 μV; normalized latency: 5.07 to 5.56 ms), indicating broadly conserved binaural processing dynamics across rodents and other mammals despite differences in auditory ecology and social organization. Our results additionally revealed significant effect of social group on BIC relative amplitude, with group-living species generally exhibiting stronger binaural responses than solitary species. However, no differences in BIC normalized latency were observed among social groups. These findings suggest that even moderate increased in social complexity are associated with enhanced sensitivity of binaural auditory processing in group-living rodents. Rodents rely on ITDs and interaural level differences (ILDs) to localize sound, which are primarily processed by neurons of the medial and lateral superior olives[73,74]. Although it remains unclear whether the differences in relative amplitude observed between group-living and solitary species are driven by variation in craniofacial features or by differences in number of neurons within the superior olivary complex, comparative studies quantifying volume of the medial nucleus of the trapezoid body (MNTB), as well as excitatory and inhibitory cell populations in the medial superior olive (MSO), and lateral superior olive (LSO) of each species would be valuable for clarifying these influences. Importantly, the strongest binaural relative amplitude responses observed in group-living species may reflect selection for more sensitive and rapid sound localization, which could provide advantages in detecting conspecific calls, avoiding predators, localizing prey, coordinating group behaviors, and maintaining group cohesion. These findings partly support our second prediction that group-living rodents would show a broader ITD range, characterized with higher amplitude binaural responses than other social groups.

The present comparative work extends our understanding of how social organization shapes auditory processing in wild-caught rodents. We showed that both group-living and monogamous rodents possess enhanced auditory sensitivity, characterized by lower ABR thresholds and click thresholds. We additionally showed that group-living species exhibit stronger BIC relative amplitude than social species, while no differences in amplitude ratio, BIC normalized latency nor interpeak latency were detected among social groups. These findings suggest that auditory systems may have evolved in concert with social complexity, potentially facilitating more efficient communication and coordination among group-members. More broadly, our results highlight a potential evolutionary link between sociality and sensory processing in rodents, confirming similar trends observed in other vertebrates. Future comparative and neuroanatomical studies, including those incorporating sex differences, ecological variables, and broader phylogenetic sampling across rodents and other tetrapods, will be essential to determine the generality of these relationships and elucidate the neural and morphological adaptations underlying them.

Methods

Animals

Experiments were conducted on 90 wild-caught male rodents, representing 9 species, captured from six locations in Oklahoma (Packsaddle wildlife management area, James Collin wildlife management area, Sandy Sanders wildlife management area, Tulsa, Selman living laboratory, and Stillwater) and one location from Kansas (Kansas University field station, Lawrence, Fig. 4). At each location, trapping was carried out along two trap-lines that consisted of twenty aluminum Sherman non-folding traps (7.62 cm × 7.62 cm × 25.4 cm) (H.B Sherman Traps, Inc., Tallahassee, FL) laid out in 5–10 m intervals between traps. The traps were supplied with whole grain oats and peanut butter. Traps were generally placed in brushes or vegetation to prevent heat stress, rain and sun exposure and were provisioned with 2-3 cotton balls for warmth from mid-December to mid-April of each year (from 2022 to 2025). Traps were set each evening at dusk (between 18:00 and 20:00 h) and checked the following morning at sunrise (between 06:00 and 09:00 h). Upon capture, rodents were placed in an unsealed plastic bag for visual identification of species and sex according to Caire[75]. Age was estimated for individuals based on body mass using information from primary literature, and the animals were divided into three age groups (juvenile, subadult, and adult, Table 1). Given the uncertainty in mating status, the variability and indirect nature of age estimation of wild-caught rodents, neither variable was included in the statistical analyses.

Animal Transportation and ethical notes

Animals were placed individually in an Innovive (Animal Specialties and Provisions, LLC, Quakertown, PA) disposable plastic mouse cage (37.34 cm length x 23.37 cm width x 13.97 cm height) equipped with food and water immediately after capture and transported to the laboratory for ABR measurements and genetic identification[26]. During this period, animals were housed individually at room temperature, and no social housing occurred between capture and testing. ABR measurements were performed immediately upon arrival in the laboratory, ~2–5 h after capture and visual identification of species, ensuring a consistent and minimal captivity duration across all individuals and species. All animal collection was

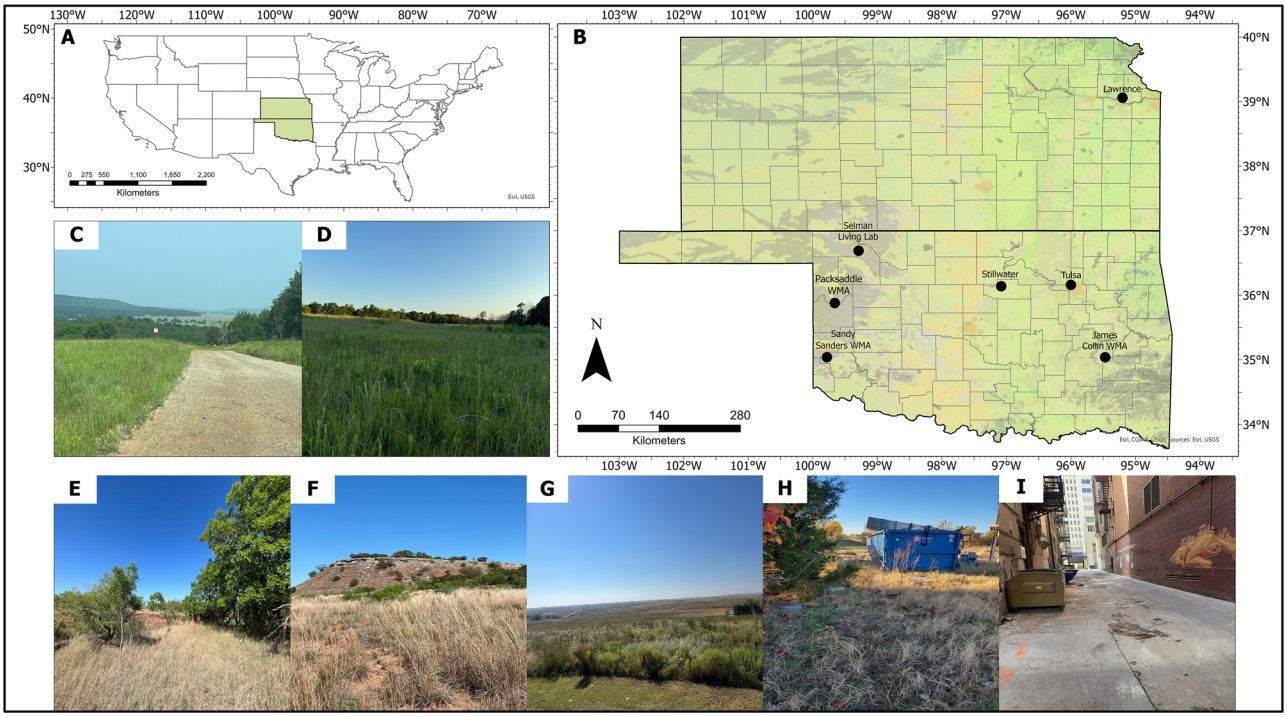


Fig. 4 | Geographic distribution of sampling sites across Oklahoma and Kansas. **A** Map of the united States, showing Oklahoma and Kansas states in green. Lines are geographical borders for each states. **B** shows each sampling location and black lines indicate geographical borders of states counties. **C**–**I** are photographs of each sites [**C** = James Collins Wildlife Management Area (WMA), **D** = Lawrence, Kansas, **E** = Packsaddle WMA, **F** = Sandy Sanders (WMA), **G** = Selman Living Lab, **H** = Stillwater, and **I** = Tulsa, all photos were taken by Luberson Joseph in the field).

authorized by the Oklahoma Department of Wildlife Conservation (ODWC) and the Kansas Department of Wildlife and Parks (KDWP). Field sampling and animal handling complied with applicable United States laws and regulations and followed the guidelines and recommendations of the American Society of Mammalogists[76], and permitted by the University of Central Oklahoma (UCO) Selman Living Laboratory (SLL) and the Kansas University (KU) Field Station. All experimental procedures involving live animals were conducted in accordance with institutional ethical guidelines and were approved by the Institutional Animal Care and Use Committee (IACUC; protocol no. 22-09).

Deoxyribonucleic acid (DNA) analysis

Tail tissue samples were used to genetically confirm species following the protocol described by Nicolas and colleagues[77]. DNA samples were primarily incubated overnight in a mixture of Buffer ALT and Proteinase K at 56 °C for cells lysing. DNA were next extracted using a Qiagen DNeasy blood and tissue extraction kit (Qiagen Inc., Hilden, Germany), following the manufacturer's guidelines. To monitor for potential DNA contamination, we performed blank readings in each extraction batch with no DNA sample (i.e., only water). We next measured the purity and concentration of the DNA extracted for each sample with a Thermo Scientific Nano-Drop (Fisher Scientific, Spectrophotometer, Nanodrop Life 6 V 18 W, Wilmington, DE). For all samples, the primer sequences (CCTACTCRGCCATTTTACCTATG) and (ACTTCTGGGTGCCAAAGAATCA) were used for amplifying the cytochrome c oxidase subunit I (COI) gene[78,79]. DNA samples were assayed in a 50 µL reaction comprising 25 µL Phusion Master mix, 2.5 µL reverse primer, 2.5 µL forward primer, and 20 µL mQ water. The total polymerase chain reaction (PCR) amplifications were performed using an Eppendorf Thermal Cycler (Eppendorf 6346000021 Master cycler Nexus SX1 Thermal Cycler, California, USA) under the following criteria: initial denaturation at 95 °C for 300 s; 35 cycles of denaturation at 94 °C for 30 s, annealing at 55 °C for 40 s, and an extension step at 72 °C for 90 s; followed by a second extension at 72 °C for 300 s. Amplified double-stranded PCR products were purified and sequenced at the Oklahoma State University's Center for Genomic and Proteomic (Stillwater, Oklahoma, USA). Resulting sequencing were adjusted manually using the software 4peaks and compared with other COI sequences using the National Center for Biotechnology Information (NCBI) GenBank database to confirm species identity[80,81].

Classification of social organization

Sociality encompasses interactions among individuals that form associations across diverse contexts (i.e., shelter use, defense, and foraging), and varies across rodents species[82–84]. We classified rodents' social structure following the criteria of Lukas and Clutton-Brock[85], which categorizes species as solitary, monogamous, and group-living systems based on patterns of parental care and the number of breeding adults (Fig. 1, Table 1). Species were classified as solitary when parental care duties were provided only by females and breeding females maintained separated home ranges, despite occasionally sharing feeding or resting sites[86–88]. In this group, social interaction is limited only to mating encounters and relations between mothers and offspring. Monogamous species were defined as those in which males and females shared equal parental care duties in raising offsprings and breeding pairs occupied a common home range for extended periods without cohabiting with additional breeding adults[89,90]. Species were classified as group-living when parental care duties were provided only by females, and two or more breeding females occupied the same home range or breeding places and females tolerated each other's presence and shared all areas within their home ranges[88,91]. Species assignments to a specific social system were based on published descriptions of social organization from the primary literature (Table 1). When multiple sources were available, classifications were based on consistent patterns reported across populations, with priority given to field-based observations. Because social organizations can vary across social contexts, these classifications represent the predominant system reported for each species rather than population-specific variation.

ABR recordings and acoustic stimulations

We applied click (0.1 ms) and tone burst [4 ms in duration with 1 ms rise and 1 ms fall ramps (2 ms ± 1 ms rise/fall ramps)] sounds for ABR measurements, similar to previous publications[25,26,28,51]. In brief, ABR recordings were acquired in a sound-attenuating chamber (Noise Barriers, Lake Forest, IL, USA) and data were filtered using a second-order bandpass filter (50–3000 Hz) and averaged over a 10-12 ms window across 500–1000 stimulus repetitions. Sound stimuli were presented at an interstimulus interval of 30 ms and a standard deviation of 5 ms, at different intensities (90, 80, 70, 60, 50, 40, 30, 20, 10, 5 dB SPL), starting with the loudest sound levels. Acoustic stimuli were generated either through multi-field speakers (MF-1, Tucker-Davis Technologies, Fl, USA) for clicks and frequencies ranging from 1 to 24 kHz or electrostatic speakers (EC-1, Tucker-Davis Technologies, Fl, USA) for frequencies between 32 and 64 kHz, which were positioned at the animal's left and right ear canals for closed-field recording. Acoustic stimuli were calibrated prior to each ABR recording session using in-ear Etymotic microphones positioned within the ear canal of the sedated subject. Both speakers were calibrated using the Etymotic microphones across a range of voltages driven to the speakers for both clicks and across frequencies, allowing for consistent sound delivery across frequency range. The ABR signals were measured with subdermal needle electrodes (Viasys Healthcare, Madison, WI, USA), which were inserted under the skin (~3–5 mm) between the pinnae at the vertex (active electrode), behind the apex at the nape (reference electrode), and the right back leg (ground electrode). During the measurement, animals were intraperitoneally anesthetized with a cocktail of ketamine (60 mg/kg) and xylazine (10 mg/kg) and kept sedated with ketamine (25 mg/kg) and xylazine (12 mg/kg) dose supplements. Anesthetic depth was monitored throughout the experiments by assessing toe pinch reflex responses and respiration. Body temperature was maintained at 37 °C throughout the experiment using a temperature-controlled water pump heating pad (25.5 cm×17.8 cm; HTP-1500 Heat Therapy Pump; Kent Scientific, Torrington, CT). ABR recordings lasted approximately 120 min (per individual), after which the wild-caught animals were humanely euthanized with pentobarbital (30 mg/kg), and transcardially perfused with phosphate-buffered saline and paraformaldehyde for subsequent tissue collection and histological analyses not reported here.

ABR data processing

Data analyses followed previously established procedures[25,26,28,51]. In short, during pre-processing, ABR data were normalized relative to the pre-stimuli baseline, and the resulting ABR waveforms were statistically tested for differences across social groups. Auditory sensitivity was assessed quantitatively using ABR and click thresholds according to Brittan-Powell and Dooling[92]. We constructed audiograms (using pure tone stimuli) for each species across frequencies (1, 2, 4, 8, 16, 32, 46, 64 kHz) and intensities (90 −5 dB SPL, presented in 10 dB SPL steps, with finer 5 dB steps near threshold when needed) to determine hearing thresholds. Hearing thresholds were considered to be between the intensity at which the waveforms were no longer visible and the previous intensity at which they were visible by both left and right monaural stimuli. Click threshold was measured by gradually reducing the intensity of the click stimulus in 5–10 dB SPL increments until the ABR waveforms were no longer noticeable (Fig. 2A). Click threshold was described as the difference in intensity between the last level where ABR waveforms were observed and the level at which they disappeared (i.e., if ABR waveforms were observable at 40 dB SPL and no longer observable at 30 dB SPL, the click threshold was estimated to be 35 dB SPL for this animal).

Individual wave latencies (time to peak amplitude) and amplitudes (voltage from peak to absolute trough) were detected for wave I and wave IV of the ABR recorded at each intensity level (90–50 dB SPL) for the left and right pinnae. We next quantified interpeak latency and amplitude ratio for each species to enable robust social-groups comparisons using the average values at each tested intensity[25,26,64]. These relative measures account for the inter-individual variability (e.g., differences in body mass, overall signal amplitudes, recording conditions, electrode placement variation, etc.,), and therefore widely used to provide a consistent index of neural gain and conduction timing across individuals and species[26,64,93].

Binaural ABR measurements were also assessed as previously described[25,26,28]. In short, to produce the binaural ABR evoked responses, we presented broadband alternating polarity click stimuli simultaneously or with an ITD (closed-field presentation) at 90 dB SPL to both pinnae of the anesthetized animal. Prior to BIC quantification, we adjusted the zero baseline of each recording at each tested ITD using the RMS of the stimulus waveform to ensure consistent alignment across measurements. The binaural interaction component (BIC) of the ABR responses was used to determine species binaural ABR response abilities and was measured as the negative peak wave (DN1, Supplementary Fig. 6) at wave IV of the ABR after subtraction of the sum of the monaural ABR responses from the binaural ABR responses[29,50,69] [$BIC = ABR_{binaural} - \left\{ABR_{left} + ABR_{right}\right\}$, where $ABR_{binaural}$ represents the binaural responses and ABR_{left} and ABR_{right} represent monaural responses]. To assess how BIC varies with ITD, rodents were presented click stimuli with shifting ITDs of 0.0 to 2.0 ms in 0.5 steps. Peak latency and amplitude of DN1 at each ITD for each individual were calculated using custom Python software. Peak latency was defined as the time point corresponding to the peak of the prominent negative peak, and peak amplitude was measured relative to the pre-stimulus baseline, as previously described[25,26,28]. Because DN1 latency is shortest at 0 ITD and DN1 amplitude is highest at 0 ITD, peak latency and amplitude values at other ITDs were normalized relative to the 0 ms condition to reduce inter-individual variability to the timing and magnitude of the binaural ABR evoked responses[50]. Normalized values were calculated as: $Normalized\ latency\ shift = Latency_{ITD} - Latency_0$, and $Relative\ amplitude = \left(\frac{Amplitude_{ITD}}{Amplitude_0}\right)X100$, where $Latency_0$ and $Amplitude_0$ correspond to the DN1 latency and amplitude at 0 ms ITD, respectively. This normalization yielded BIC normalized DN1 latency shifts and BIC relative DN1 amplitude, which we used the average to compare binaural responses as a function of ITD across social groups.

Statistics and reproducibility

Data were analyzed using phylogenetic generalized linear mixed models (PGLMMs), with Bayesian posterior distributions estimated via Markov Chain Monte Carlo (MCMC) sampling to account for phylogenetic relatedness among species. We obtained a species-level time-calibrated phylogenetic tree by pruning the phylogeny of Upham and colleagues[94] using the R package APE[95] to include only the species represented in this study. Separate models were fitted for each response variable (frequency thresholds, click thresholds, interpeak latency, amplitude ratio, BIC normalized DN1 latency and BIC relative DN1 amplitude), all treated as Gaussian response variables. Fixed effects included sociality (solitary, monogamous, group-living), intensity (90–50 dB SPL), frequency (1-64 kHz), and ITD (0.0–2.0 ms), as well as their interaction terms where specified. Frequency, intensity, and ITD were treated as categorical variables to reflect the discrete stimulus conditions used in the experimental design. To account for the repeated measures structure of our data, both phylogenetic relationships among species and individual identity (animal ID) were modeled as random effects. Inclusion of animal ID as random intercept accounts for non-independence of repeated measurements recorded from the same individuals across frequencies, intensities, or ITDs, while phylogenetic random effect accounts for shared evolutionary history among species. Phylogenetic signal was quantified using phylogenetic heritability (H^2), calculated as $H^2 = \frac{\sigma^2 a}{(\sigma^2 a + \sigma^2 e)}$, where $\sigma^2 a$ represents the phylogenetic variance and $\sigma^2 e$ represents the residual (non-phylogenetic) variance from the best supported models[96]. Because our central hypothesis focuses on evolutionary relationships between sociality and auditory traits, species was treated as the primary unit of inference, while individual-level measurements ($n = 90$) contributed to species-level estimates across models. For the MCMC sampling, we used the default diffuse Gaussian prior in MCMCglmm for fixed

effects (mean = 0, variance = 10^{10}) and a weakly informative inverse-Gamma prior ($V = 1$) for random effects with shape and scale parameters set to 0.001. Each model was run for 260,000 interactions, with a burn-in of 10,000 and thinning every 250 steps to reduce autocorrelation in the chains. Three independent chains were run for each model, and we used the German-Rubin test to confirm model convergence[97–99], with scale reduction factors of ~1 indicating convergence across all models.

To select the model that best fit each response variable, we compared four candidate models: (i) a null model including random effects (phylogeny and animal ID) and a single predictor (frequency, intensity, or ITD), (ii) an additive model including random effects and two predictors (sociality + frequency, sociality + intensity, or sociality + ITD), (iii) a full model including random effects, and interaction terms (sociality * frequency, sociality * intensity, or sociality * ITD), and (iv) a full model with body mass (BM) as a covariate to account for interspecific size differences. Model fit was evaluated using Deviance Information Criterion (DIC; Supplementary Table 3). When differences in DIC score across models were small ($\Delta DIC < 2$), the more parsimonious model was selected. In most cases, the full model was best supported, except for the amplitude ratio and interpeak latency, where the additive model yields lower DIC values. All comparative analyses were conducted within a Bayesian framework. Effect sizes are reported as posterior means with 95% confidence intervals. Fixed effects were denoted statistically significant when 95% CIs did not overlap zero and when P_{MCMC} value was less than 0.05. To test for differences in frequency, intensity, or ITD among social groups, we performed a Bayesian pairwise contrast test (two-tailed comparisons of posterior distributions). All PGLMM models were fitted using the MCMCglmm package[99,100]. All figures were generated in Python and R (version 4.0.3)[101], using the "ggplot2" packages[102].

Data availability

Source data underlying figures in the main manuscript and Supplementary Figs. can be found at[103,104].

Code availability

All codes used to create the figures and statistics have been deposited in fig share and can be downloaded at https://doi.org/10.6084/m9.figshare.30827288.

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